niconieves — ubuntu@ip-172-31-35-135: ~ — ssh -i CST4715_Spring26.pem ubuntu@ec2-3-80-242-148.compute-1.amazonaws.com — 202×55

```
web-7bcc44886d-pbkc7   1/1     Running   0          89s
web-7bcc44886d-sq88k   1/1     Running   0          30s
ubuntu@ip-172-31-35-135:~$ cat > svc-clusterip.yaml << 'EOF'
apiVersion: v1
kind: Service
metadata:
  name: web-svc
spec:
  selector:
    app: web
  ports:
  - port: 80
    targetPort: 80
  type: ClusterIP
EOF
ubuntu@ip-172-31-35-135:~$ kubectl apply -f svc-clusterip.yaml
kubectl get svc web-svc
service/web-svc created
NAME      TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
web-svc   ClusterIP   10.99.41.157   <none>        80/TCP    0s
ubuntu@ip-172-31-35-135:~$ kubectl describe svc web-svc | grep Endpoints
Endpoints:                10.244.0.16:80,10.244.0.15:80,10.244.0.14:80
ubuntu@ip-172-31-35-135:~$ kubectl run test-pod --image=curlimages/curl --restart=Never --rm -it -- curl web-svc
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
pod "test-pod" deleted from default namespace
ubuntu@ip-172-31-35-135:~$ kubectl delete svc web-svc
kubectl delete deployment web
service "web-svc" deleted from default namespace
deployment.apps "web" deleted from default namespace
ubuntu@ip-172-31-35-135:~$ kubectl get all
NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   18m
ubuntu@ip-172-31-35-135:~$
```

niconieves — ubuntu@ip-172-31-35-135: ~ — ssh -i CST4715_Spring26.pem ubuntu@ec2-3-80-242-148.compute-1.amazonaws.com — 202×55

```
 Type    Reason             Age   From                   Message
 ----    ------             ----  ----                   -------
 Normal  ScalingReplicaSet  92s   deployment-controller  Scaled up replica set web-7bcc44886d from 0 to 3
ubuntu@ip-172-31-35-135:~$ kubectl scale deployment/web --replicas=5
ubectl get pods -l app=web
eployment.apps/web scaled
AME                    READY   STATUS              RESTARTS   AGE
eb-7bcc44886d-cbflp    1/1     Running             0          2m19s
eb-7bcc44886d-gx2q5    0/1     Pending             0          0s
eb-7bcc44886d-pw78p    0/1     ContainerCreating   0          0s
eb-7bcc44886d-txwtj    1/1     Running             0          2m19s
eb-7bcc44886d-vvktx    1/1     Running             0          2m19s
buntu@ip-172-31-35-135:~$ kubectl get pods -l app=web
AME                    READY   STATUS    RESTARTS   AGE
eb-7bcc44886d-cbflp    1/1     Running   0          2m58s
eb-7bcc44886d-gx2q5    1/1     Running   0          39s
eb-7bcc44886d-pw78p    1/1     Running   0          39s
eb-7bcc44886d-txwtj    1/1     Running   0          2m58s
eb-7bcc44886d-vvktx    1/1     Running   0          2m58s
buntu@ip-172-31-35-135:~$ kubectl scale deployment/web --replicas=2
ubectl get pods -l app=web
eployment.apps/web scaled
AME                    READY   STATUS        RESTARTS   AGE
eb-7bcc44886d-cbflp    1/1     Running       0          3m29s
eb-7bcc44886d-gx2q5    1/1     Terminating   0          70s
eb-7bcc44886d-pw78p    0/1     Completed     0          70s
eb-7bcc44886d-txwtj    1/1     Running       0          3m29s
eb-7bcc44886d-vvktx    1/1     Terminating   0          3m29s
buntu@ip-172-31-35-135:~$ kubectl get pods -l app=web
AME                    READY   STATUS    RESTARTS   AGE
eb-7bcc44886d-cbflp    1/1     Running   0          4m4s
eb-7bcc44886d-txwtj    1/1     Running   0          4m4s
buntu@ip-172-31-35-135:~$ kubectl rollout history deployment/web
eployment.apps/web
EVISION  CHANGE-CAUSE
         <none>

buntu@ip-172-31-35-135:~$ kubectl set image deployment/web nginx=nginx:1.25-alpine
eployment.apps/web image updated
buntu@ip-172-31-35-135:~$ kubectl rollout status deployment/web
eployment "web" successfully rolled out
buntu@ip-172-31-35-135:~$ kubectl get replicasets
AME              DESIRED   CURRENT   READY   AGE
eb-7bcc44886d    0         0         0       5m20s
eb-7cfddf6b6f    2         2         2       18s
buntu@ip-172-31-35-135:~$ kubectl describe pods -l app=web | grep Image:
   Image:         nginx:1.25-alpine
   Image:         nginx:1.25-alpine
buntu@ip-172-31-35-135:~$ kubectl rollout history deployment/web
eployment.apps/web
EVISION  CHANGE-CAUSE
         <none>
         <none>

buntu@ip-172-31-35-135:~$
```

niconieves — ubuntu@ip-172-31-35-135: ~ — ssh -i CST4715_Spring26.pem ubuntu@ec2-3-80-242-148.compute-1.amazonaws.com — 202×55

```
kubeconfig: Configured

NAME       STATUS   ROLES           AGE    VERSION
minikube   Ready    control-plane   119s   v1.35.1
ubuntu@ip-172-31-35-135:~$ cat > rs.yaml << 'EOF'
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: web-rs
  labels:
    app: web
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: nginx:alpine
        ports:
        - containerPort: 80
EOF
ubuntu@ip-172-31-35-135:~$ kubectl apply -f rs.yaml
kubectl get pods -l app=web
replicaset.apps/web-rs created
NAME           READY   STATUS              RESTARTS   AGE
web-rs-967qs   0/1     ContainerCreating   0          0s
web-rs-c5djv   0/1     ContainerCreating   0          0s
web-rs-c5l28   0/1     ContainerCreating   0          0s
ubuntu@ip-172-31-35-135:~$ kubectl get pods -l app=web
NAME           READY   STATUS    RESTARTS   AGE
web-rs-967qs   1/1     Running   0          43s
web-rs-c5djv   1/1     Running   0          43s
web-rs-c5l28   1/1     Running   0          43s
ubuntu@ip-172-31-35-135:~$ kubectl delete pod $(kubectl get pods -l app=web -o name | head -1)
error: there is no need to specify a resource type as a separate argument when passing arguments in resource/name form (e.g. 'kubectl get resource/<resource_name>' instead of 'kubectl get resource resou
rce/<resource_name>'
ubuntu@ip-172-31-35-135:~$ kubectl get pods -l app=web
NAME           READY   STATUS    RESTARTS   AGE
web-rs-967qs   1/1     Running   0          115s
web-rs-c5djv   1/1     Running   0          115s
web-rs-c5l28   1/1     Running   0          115s
ubuntu@ip-172-31-35-135:~$ kubectl delete pod $(kubectl get pods -l app=web -o name | head -1 | cut -d/ -f2)
pod "web-rs-967qs" deleted from default namespace
ubuntu@ip-172-31-35-135:~$ kubectl get pods -l app=web
NAME           READY   STATUS    RESTARTS   AGE
web-rs-c5djv   1/1     Running   0          2m23s
web-rs-c5l28   1/1     Running   0          2m23s
web-rs-jrgn5   1/1     Running   0          4s
ubuntu@ip-172-31-35-135:~$
```

niconieves — ubuntu@ip-172-31-35-135: ~ — ssh -i CST4715_Spring26.pem ubuntu@ec2-3-80-242-148.compute-1.amazonaws.com — 202×55

```
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat May  9 03:50:18 UTC 2026

  System load:  0.03               Temperature:           -273.1 C
  Usage of /:   11.4% of 23.17GB   Processes:             119
  Memory usage: 13%                Users logged in:       0
  Swap usage:   0%                 IPv4 address for ens5: 172.31.35.135


Expanded Security Maintenance for Applications is not enabled.

30 updates can be applied immediately.
27 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


Last login: Sat May  9 03:36:16 2026 from 71.183.199.245
ubuntu@ip-172-31-35-135:~$ minikube start --driver=docker
    minikube v1.38.1 on Ubuntu 24.04
    Using the docker driver based on user configuration
!   Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.

    The requested memory allocation of 1907MiB does not leave room for system overhead (total system memory: 1907MiB). You may face stability issues.
    Suggestion: Start minikube with less memory allocated: 'minikube start --memory=1907mb'

    Using Docker driver with root privileges
    Starting "minikube" primary control-plane node in "minikube" cluster
    Pulling base image v0.0.50 ...
    Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 236.67
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 125.51
    Creating docker container (CPUs=2, Memory=1907MB) ...
    Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
    Configuring bridge CNI (Container Networking Interface) ...
    Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
    Enabled addons: storage-provisioner, default-storageclass
    Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@ip-172-31-35-135:~$ minikube status
kubectl get nodes
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

NAME       STATUS   ROLES           AGE    VERSION
minikube   Ready    control-plane   119s   v1.35.1
ubuntu@ip-172-31-35-135:~$
```